

CPR Esoft Quiz 2

Question #: 1

A 20-year-old man comes to the physician for a routine examination. He has no significant past medical history and takes no medications. Physical examination shows eruptive xanthomas. Fasting blood samples are obtained for lipid studies. The fasting serum sample appears milky and turbid. A creamy layer is observed after centrifugation. Results of laboratory studies are shown in the table. Which of the following proteins is most likely deficient in this patient?

- A. Apolipoprotein E
- B. Apolipoprotein B100
- C. LDL receptor
- D. Lipoprotein lipase (LPL)
- E. Lecithin cholesterol acyltransferase (LCAT)
- F. Cholesterol-ester transfer protein (CETP)
- G. Microsomal triglyceride transfer protein (MTP)

Attachment:

Test	Result	Reference Range
Total cholesterol	180 mg/dL	<200 mg/dL
Triglycerides	900 mg/dL	<150 mg/dL
HDL cholesterol	40 mg/dL	>40 mg/dL
LDL cholesterol	90 mg/dL	<130 mg/dL

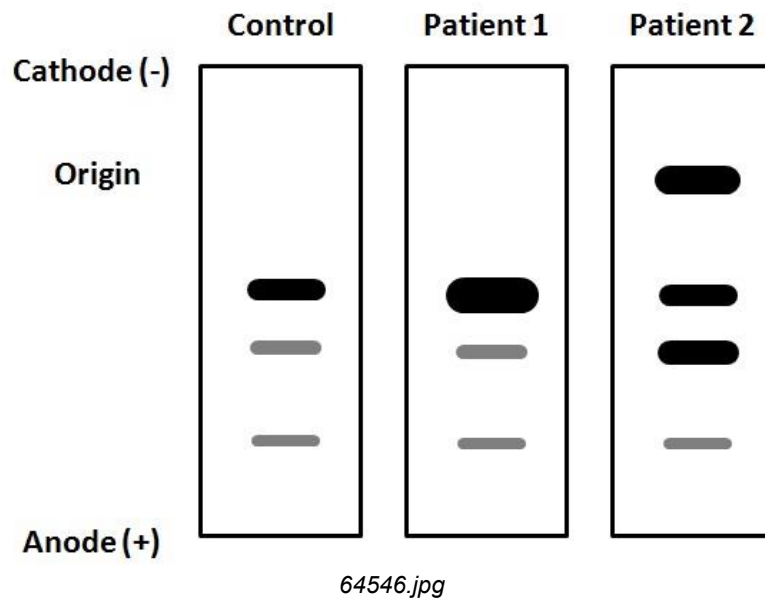
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Question #: 2

Fasting serum lipoprotein electrophoresis is performed on samples from two patients with lipoprotein disorders and a normal control sample. The results of the test are shown. Which of the following is the best interpretation of these laboratory findings?

- A. Patient 1 has Type IIB hyperlipoproteinemia
- B. Patient 2 has Type V hyperlipoproteinemia
- C. Patient 1 would have high serum triglyceride levels
- D. Patient 1 has a mutation in the ApoE (remnant) receptor
- E. Patient 2 would have high serum total cholesterol and high LDL cholesterol

Attachment:



Question #: 3

A 40-year-old man comes to the physician because of a 2-day history of tightness in the chest while walking uphill. His father died of a myocardial infarction at the age of 55 years. His blood pressure is 160/95 mmHg. Physical examination shows palmar xanthomas. Results of fasting laboratory studies are shown in the table. Coronary angiography shows atherosclerotic narrowing of multiple coronary arteries. The patient's condition is most likely caused by a defect in which of the following processes?

- A. Receptor mediated endocytosis by Apo E
- B. Synthesis of Apo B-48
- C. Mutations in the LDL receptor gene
- D. Expression of lipoprotein lipase
- E. Production of lecithin cholesterol acyltransferase

Attachment:

Test	Result	Reference Range
Total cholesterol	340 mg/dL	<200 mg/dL
Triglycerides (TAG)	300 mg/dL	<150 mg/dL
HDL cholesterol	38 mg/dL	>40 mg/dL

92788.jpg

Question #: 4

A 35-year-old man is brought to the emergency department because of a sudden onset of crushing substernal chest pain that radiates to his neck. He has had prior episodes of exertional chest pain relieved by rest. His father died at the age of 40 years from a myocardial infarction. Fasting serum

lipoprotein electrophoresis shows prominent Beta and Pre-Beta bands, with a decreased alpha lipoprotein band. Which of the following is the most likely lipid disorder in this patient?

- A. Type IIA (Familial hypercholesterolemia)
 - B. Type IIB (Familial combined hyperlipidemia)
 - C. Type III (Familial dysbetalipoproteinemia)
 - D. Type I hypertriglyceridemia
 - E. Type IV hypertriglyceridemia
 - F. Hypoalphalipoproteinemia
 - G. Hypobetalipoproteinemia
-

Question #: 5

A 3-month-old boy is brought to the physician by his mother because of recurrent vomiting, diarrhea, and poor weight gain since birth. Physical examination shows abdominal distension. Laboratory studies show increased fecal fat and steatorrhea. The results of the fasting serum lipid profile are shown in the table. He is prescribed a specialized diet that includes supplementation with vitamins A, D, E, and K. Which of the following proteins is most likely deficient in this patient?

- A. Membrane ABCA1-cholesterol transporter
- B. Apolipoprotein A-I
- C. Apolipoprotein C-II
- D. Cholesteryl ester transfer protein (CETP)
- E. Microsomal triglyceride transfer protein (MTP)
- F. Lecithin cholesterol acyl transferase (LCAT)

Attachment:

Test	Result	Reference Range
Triglycerides	15 mg/dL	<150 mg/dL
Total cholesterol	40 mg/dL	<200 mg/dL

311715.jpg

Question #: 6

A 60-year-old man comes to the physician for a follow-up evaluation. He has a 3-year history of type 2 diabetes mellitus and hypertension. Six months ago, he began taking his medications regularly after experiencing a myocardial infarction. He reports that since then, he has made several lifestyle changes: he reduced red meat intake, walks for 30 minutes daily, quit smoking, and attends regular follow-up visits every 2 months. His blood pressure measurements over the past several visits have ranged from 136–140/80–88 mm Hg.

Results of laboratory studies are shown:

Hemoglobin A1c: 6.5% (Reference: 5.7-6.4%)

Serum triglycerides: 180 mg/dL (Reference: <150 mg/ dL)

HDL-cholesterol: 30 mg/dl (Reference: >40 mg/ dL).

Dietary supplementation of which of the following nutrients will most likely positively affect his HDL levels?

- A. Trans fatty acids
 - B. Palmitic acid
 - C. Oleic acid
 - D. Alpha-linolenic acid
 - E. Folic acid
 - F. Ascorbic acid
-

Question #: 7

An investigator is studying tissue-specific processing of the apolipoprotein B (ApoB) pre-mRNA. It is observed that an enzymatic modification of a single base results in different ApoB mRNAs being produced in the liver and intestinal mucosal cells. In the liver, the full-length ApoB-100 protein is synthesized, whereas in the intestinal mucosal cells, a shorter ApoB-48 protein is produced from the same gene. Which of the following mechanisms is most likely responsible for producing this diversity of proteins in these cells?

- A. RNA editing
 - B. Alternative splicing
 - C. Poly-adenylation
 - D. Proteolytic cleavage
 - E. RNA capping
-

Question #: 8

A 56-year-old man is brought to the emergency room because of a 2-hour history of shortness of breath, nausea, and palpitations. His pulse is 132/min, respirations are 18/min, and his blood pressure is 162/96 mmHg. Cardiac auscultation shows an irregular heartbeat, varying from 120-140 bpm. ECG confirms atrial fibrillation. Administration of an experimental drug that inhibits the T-type calcium channels, lowers his heart rate to 92/min. There is no change in cardiac contractility or diastolic aortic pressure. This drug is most likely to cause which of the following effects on the cardiac action potentials?

- A. A
- B. B
- C. C

D. D

E. E

Attachment:

	SA node affected phase	SA node depolarization rate
A	Phase 4	Decreased
B	Phase 0	Increased
C	Phase 4	Increased
D	Phase 0	Decreased
E	Phase 3	Increased

Capture.PNG

Question #: 9

An 83-year-old man comes to the physician because of a 1-week history of shortness of breath. He has a 20-year history of hypertension and a 10-year history of coronary artery disease. His pulse is 100/min, blood pressure is 164/94 mm Hg, and respirations are 20/min. Physical examination shows elevated jugular venous pulse, pitting edema of the ankles and bilateral basal lung crepitation. He is diagnosed with congestive heart failure. Cardiac catheterization study shows:

Systemic arterial oxygen content = 0.2 mL O₂/mL,

Pulmonary arterial oxygen content = 0.1 mL O₂/mL and

Total oxygen consumption = 400 mL O₂/min.

Using the above data, which of the following values best represents this patient's stroke volume?

A. 30 mL/beat

B. 40 mL/beat

C. 50 mL/beat

D. 60 mL/beat

E. 70 mL/beat

Question #: 10

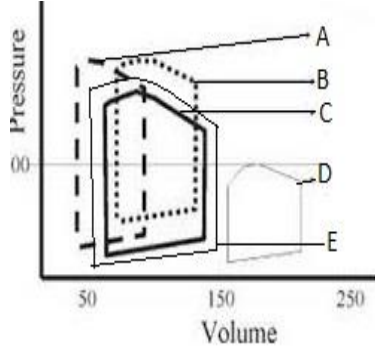
A 62-year-old woman comes to the physician because of weakness, fatigue, fever, and weight gain 5 years following her radiation therapy post breast cancer surgery. She also complains about abdominal discomfort and difficulty breathing doing work. Physical examination shows hepatomegaly and jugular venous distention that fails to subside on inspiration but shows no evidence of hypotension or pulsus paradoxus. An echocardiogram shows reduced end-diastolic volumes, elevated diastolic pressures in both ventricles and a slightly elevated end-systolic volume. Which of the following pressure-volume loops will most likely be associated with this patient?

A. A

B. B

- C. C
- D. D
- E. E

Attachment:



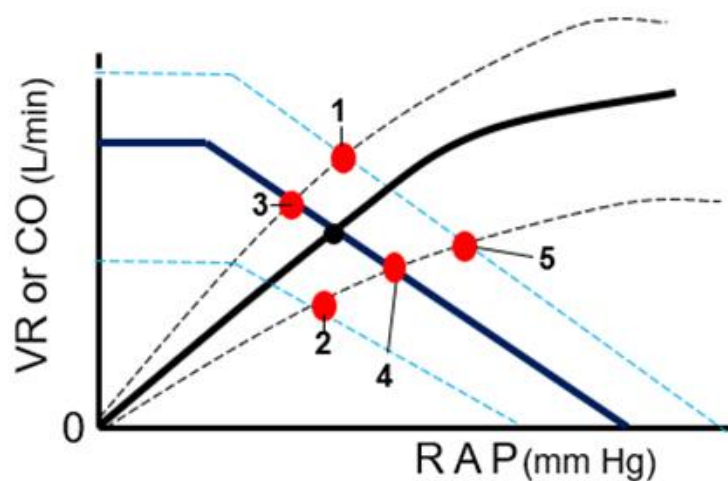
PV LOOPS.jpg

Question #: 11

A 65-year-old woman is brought to the emergency department because of worsening shortness of breath and swelling in both feet. She has a 3-month history of decompensated heart failure. Which of the labeled points on the attached Guyton cross plot most likely reflects compensated heart failure?

- A. 1
- B. 2
- C. 3
- D. 4
- E. 5

Attachment:



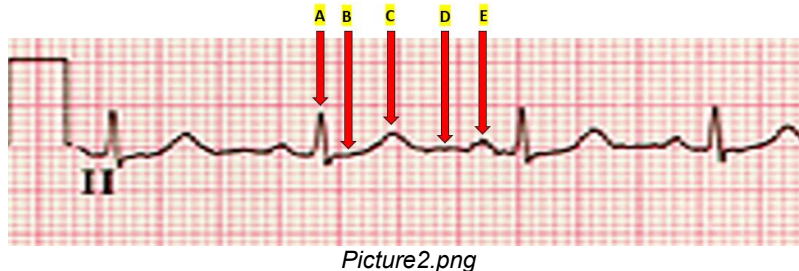
CrossPlot.png

Question #: 12

A 35-year-old man comes to the physician for a routine examination. He has been part of the national cycling team for 15 years. His blood pressure is 126/76 mmHg, pulse is 61/min and respirations are 12/min. Cardiac examination shows a sustained apical impulse that is laterally displaced. Echocardiography shows increased thickness of the left ventricular walls. An ECG recording is shown in the attached figure. Which of the following labeled components of the ECG most likely represents electrical depolarization of the thickened ventricle in this patient?

- A. A
- B. B
- C. C
- D. D
- E. E

Attachment:

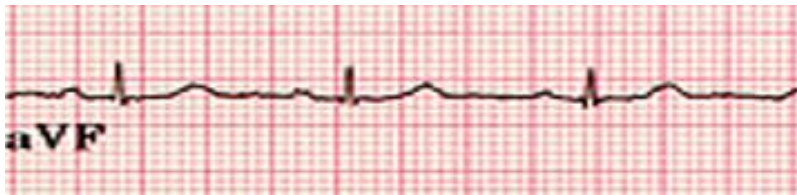


Question #: 13

A 61-year-old woman comes to the physician because of a 2-day history of chest discomfort. She has a 12-year history of migraines and uses a beta-blocker. Her pulse is 82/min, blood pressure is 138/82 mmHg, and respirations are 12/min. Physical examination shows no abnormalities. An ECG is shown in the attached figure. Which of the following lead placements most likely corresponds to the recording from the lead shown in the attached figure?

- A. Positive electrode located in the left leg
- B. Positive electrode located in the left shoulder
- C. Positive electrode located in the right shoulder
- D. Negative electrode located in the left leg
- E. Negative electrode located in the left shoulder

Attachment:



Picture6.png

Question #: 14

A 20-year-old man comes to the physician because of a 1-week history of fever, persistent cough, chest pain, and shortness of breath. He is diagnosed with bronchopneumonia and treated with a 1-week course of antibiotics. An ECG recorded in leads I and II are shown in the attached figure. Which of the following is the most likely value of the mean electrical axis of this patient's heart?

- A. $+0^\circ$
- B. -30°
- C. $+30^\circ$
- D. $+60^\circ$
- E. $+90^\circ$

Attachment:



ECG4.jpg

Question #: 15

A 45-year-old man is brought to the emergency department because of a 2-hour history of palpitations. He says that he is feeling "flutters" in his chest. He has a 10-year history of poorly controlled high blood pressure. His pulse is 104/min and has an irregular rhythm, blood pressure is 146/86 mmHg, and respirations are 18/min. Physical examination shows a loud S2 in the aortic area. An ECG shows atrial

flutter. The net QRS amplitudes measured in the leads I and aVF show:

Lead I =2 mV and

Lead aVF =3 mV.

Which of the following leads is most likely to show the greatest net positive QRS amplitude?

- A. Lead I
 - B. Lead II
 - C. Lead III
 - D. Lead aVR
 - E. Lead aVL
-

Question #: 16

A 65-year-old man is brought to the emergency department because of a sudden onset of chest pain and shortness of breath. He has hypertension and type 2 diabetes mellitus. ECG reports show ST-elevation in anterolateral leads. Which of the following physiological changes is most likely present in this patient?

- A. Decreased systemic vascular resistance
 - B. Increased cardiac contractility
 - C. Decreased pulmonary capillary wedge pressure
 - D. Increased end-diastolic LV pressure
 - E. Decreased pulmonary artery systolic pressure
-

Question #: 17

A 68-year-old woman is brought to the emergency department because of a 2-hour history of shortness of breath and palpitations. She has no family history of heart disease. Her pulse ranges from 86-125/min, blood pressure is 142/76 mmHg, and respirations are 18/min. Physical examination shows an irregularly irregular pulse. An ECG shows atrial fibrillation. She is administered a Class III antiarrhythmic drug that blocks the delayed rectifier potassium channels in the cardiac myocytes which slow the process of repolarization. Which of the following ECG changes is most likely to be caused by this drug?

- A. Prolonged PR interval
- B. Widened QRS complex
- C. Irregular RR interval

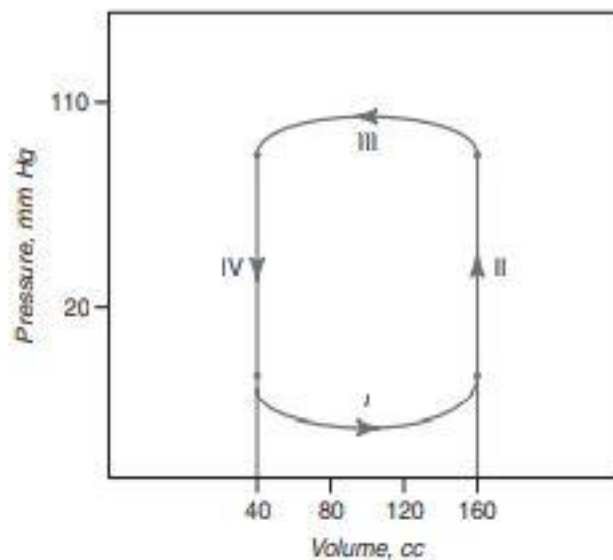
- D. Prolonged QT interval
- E. Decreased ST segment

Question #: 18

A 54-year-old woman comes to the physician because of a 3-week history of persistent general malaise and fever. The symptoms developed slowly over the weeks following her root canal treatment 3 months ago. Her temperature is 38.3°C (101°F). On cardiac examination, a click and a systolic murmur are auscultated in the mitral area. She tells the physician that the click is due to a mechanical valve replacement done 4 years ago due to rheumatic fever as a child. The phases of a cardiac cycle is shown in the attached graph. Which of the following is the most likely phase of the cardiac cycle during which the murmur is heard in this patient?

- A. Phase I
- B. Phase II
- C. Phase III
- D. Phase IV
- E. Not seen on this graph

Attachment:



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cardiac cycle.JPG

Question #: 19

A 76-year-old woman comes to the physician for a pre-operative evaluation for knee surgery. Her blood

pressure is 200/120 mm Hg. While in the hospital room, she sings in order to relax, and her blood pressure drops to 180/90 mm Hg. Using the attached image, which of the following vascular changes is most likely responsible for decreasing the patient's blood pressure?

- A. A
- B. B
- C. C
- D. D
- E. E

Attachment:

	Total peripheral resistance	Venous return
A	Increased	Increased
B	Decreased	Decreased
C	Increased	Decreased
D	Decreased	Increased
E	Increased	No change

CPRHemo.jpg

Question #: 20

An 18-year-old man comes to the physician for a routine examination. He has a family history of heart disease. He is very athletic and has been training for running competitions for several years. An ECG shows left axis deviation. An echocardiogram shows bradycardia, low but normal ejection fraction, and a dilated left ventricle with a thickened wall. There is no evidence of hypertension or valve disease. Which of the following hemodynamic changes is most likely in this patient?

- A. Increased systolic pressure

- B. Increased transmural pressure
 - C. Increased wall tension
 - D. Increased diastolic pressure
 - E. Decreased radius
-

Question #: 21

A 40-year-old woman volunteers to participate in a study examining the physiological effects of breath holding. While sitting, she holds her breath for 2 minutes (prolonged phase 2 Valsalva). During this apnea, her pulse increases from 60 to 76/min, mean arterial pressure decreases, and stroke volume decreases from 100 ml to 50 ml. The reduction in the patient's stroke volume during the apnea most likely results from a decrease in which of the following parameters?

- A. Venous compression
 - B. Intrachamber pressures
 - C. Cardiac compression
 - D. Cardiac filling
-

Question #: 22

A 50-year-old woman comes to the physician because of a 2-month history of hypertension. She has a strong family history of coronary artery disease. Her blood pressure has ranged from 150/95mmHg to 130/85mmHg during the previous three visits within the past two months. Her blood pressure today is 140/90mmHg. Physical examination shows no other abnormalities. Which of the following values best approximates her mean arterial pressure today?

- A. 119 mm Hg
 - B. 113 mm Hg
 - C. 107 mm Hg
 - D. 101 mm Hg
 - E. 95 mm Hg
-

Question #: 23

During an experiment, four different vessels of the same length are sampled. An indicator dye is mixed with blood and injected through the vessel. A decrease in which of the following would most likely be

expected to increase the blood flow in the vessel?

- A. Plasma sodium concentration
 - B. Plasma oncotic pressure
 - C. Radius of the vessel
 - D. Pressure gradient across the vessel
 - E. Viscosity of the blood
-

Question #: 24

48-year-old obese man comes to the physician because of a 10-month history of lower leg pain that occurs after walking long distances. It is relieved with rest. He has a 10-year history of hyperlipidemia. His blood pressure is 185/85 mm Hg, pulse is 83/min, and his respiratory rate is 18/min. Doppler studies of the lower limbs show increased arterial pressure in the major feeding arteries. Examination of a biopsy specimen of these arteries shows arteriosclerosis. Which of the following physiological changes in the affected arteries is most likely the cause of elevated pressure in this patient?

- A. increased laminar flow
 - B. increased elasticity
 - C. increased radius
 - D. decreased compliance
 - E. decreased tension
-

Question #: 25

A study was conducted to determine the hemodynamic effect of a drug on chronic heart failure. The participants of the study included patients with decreased contractile force and mechanically impaired left ventricle filling. The results of the study showed that the drug had a positive inotropic effect. Which of the following drugs was most likely used in the study?

- A. Beta receptor antagonist
 - B. Cardiac glycoside
 - C. Diuretic
 - D. ACE inhibitor
 - E. Angiotensin II receptor antagonist
-

Question #: 26

A 77-year-old man comes to the physician because of a 6-month history of swelling of his legs and feet. He has a 40-year history of alcoholism and a 5-year history of hepatic disease. Physical examination shows ascites and a 2+ edema of the lower extremities. A decrease in which of the following most likely promotes edema formation in this patient?

- A. Capillary hydrostatic pressure
 - B. Filtration coefficient
 - C. Interstitial colloid osmotic pressure
 - D. Interstitial fluid hydrostatic pressure
 - E. Plasma colloid oncotic pressure
-

Question #: 27

A 69-year-old man comes to the physician because of a 2-day history of black, tarry stools. He has a 30-year history of alcoholism and liver disease. His pulse is 82/min, blood pressure is 142/86 mmHg, and respirations are 16/min. Physical examination shows abdominal distension. Abdominal ultrasound shows the presence of ascites. If the values that determine the capillary fluid exchange are as follows:

Hydrostatic capillary pressure = 21 mmHg

Capillary oncotic pressure = 6 mmHg

Tissue interstitial pressure = 10 mmHg

Tissue oncotic pressure = 6 mmHg

Which of the following values best approximates the net driving force?

- A. - 10 mmHg
 - B. + 11 mmHg
 - C. + 14 mmHg
 - D. + 16 mmHg
 - E. + 22 mmHg
-

Question #: 28

A 64-year-old man comes to the physician because of a 2-day history of cough and shortness of breath. He has a 20-year history of high blood pressure. His pulse is 108/min, blood pressure is 168/92 mmHg, and respirations are 16/min. Auscultation of the lungs shows crackles in both lung fields. An x-ray of the chest shows bronchopneumonia. The patient is admitted to the ward and intravenous infusion with isotonic fluid is started. Which of the following changes in the venous return curve is most likely a result of the intravenous infusion?

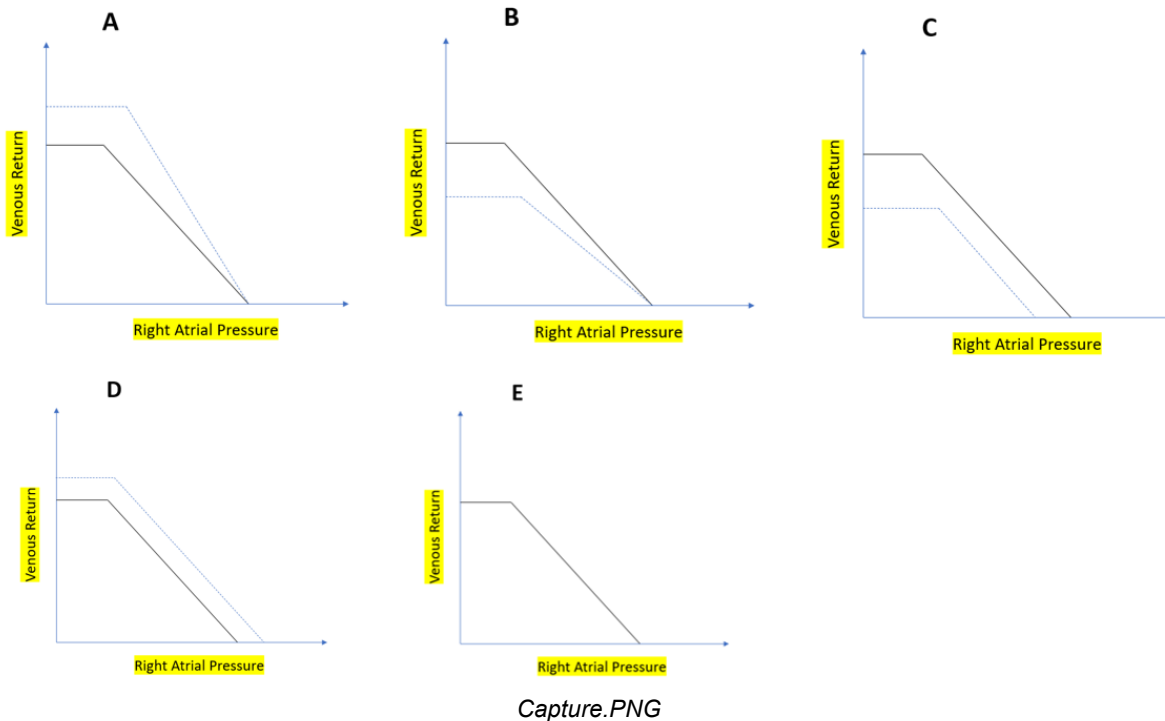
- A. A
- B. B
- C. C

D. D

E. E

Attachment:

Solid line: normal
Dotted line: patient



Question #: 29

A 45-year-old man comes to the emergency room because of chest pains for the past 3-weeks which worsens on deep inspiration. His pulse is 88/min, his respirations are 14/min, and his blood pressure is 120/80 mmHg. His Guyton cross plot is shown in the attached image. Which of the following labeled points best represents his mean systemic filling pressure?

A. A

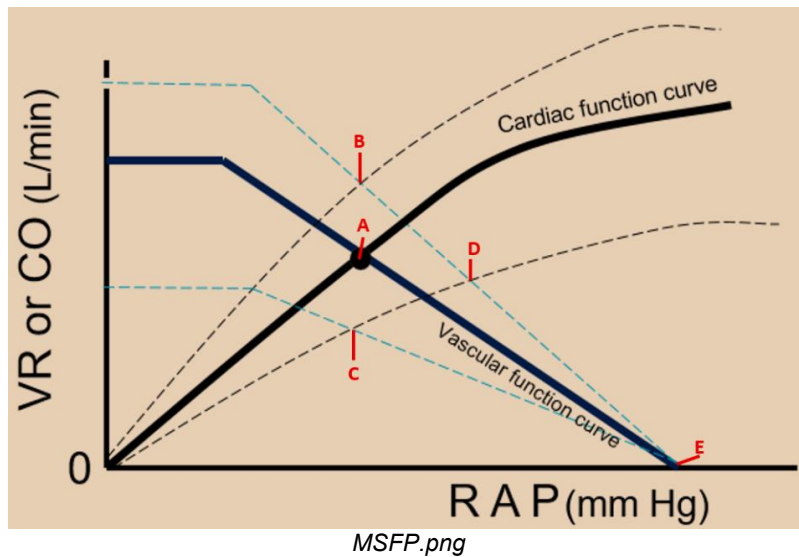
B. B

C. C

D. D

E. E

Attachment:



Question #: 30

A 71-year-old man comes to the physician because of a 3-day history of pain and swelling in his left leg. He underwent an operation for intestinal occlusion 5 weeks ago. Physical examination shows marked swelling of the leg with dilated superficial veins. Doppler ultrasound shows an extensive left leg deep venous thrombosis, with complete occlusion. Anticoagulation and thrombolytic treatments are commenced. Following treatment, which of the following mechanisms most likely controls local blood flow to the skeletal muscle distal to the occlusion?

- A. Active hyperemia
- B. Reactive hyperemia
- C. Myogenic response
- D. Prostacyclin mediated vasodilation
- E. Sympathetic mediated vascular tone